

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

1. Problem-solving with Lorentz transformations.

- (a) (40 points) Solve Morin exercise 1.34 (p. 51) *exclusively* using Lorentz transformations; in other words, don't explicitly relate any lengths or times using length contraction or time dilation, since these will naturally be encoded in the Lorentz transformations. Solve the problem for a general car velocity v . You should find that there is a particular value of v for which the tree is at the same location in the ground frame as the eastward person: what is that velocity? *Hint – there are 4 spacetime events: eastern person claps, western person claps, car passes western person, car passes tree. You will make your life easier by synchronizing the origins of the two coordinate systems to one of these events.*

A systematic way to approach this problem is to define 4 spacetime events and give their coordinates in frame S (the ground frame) and frame S' (the frame moving with the car):

Label	Event	Coords in S	Coords in S'
A	eastern person claps	(t_A, x_A)	(t'_A, x'_A)
B	western person claps	(t_B, x_B)	(t'_B, x'_B)
C	car passes western person	(t_C, x_C)	(t'_C, x'_C)
D	car passes tree	(t_D, x_D)	(t'_D, x'_D)

Our goal is to find x_D , the location in the ground frame when the car passes the tree. We can feel free to synchronize the origins of S and S' at any of the events: in these solutions we will choose to make A the origin, but any other choice will be equally valid. The information in the problem tells us that $t'_A = t'_C$, but we have set the origin of S at event A so $t'_A = t'_C = 0$. Likewise, $t'_D = t'_B$. Since the car is moving at the same velocity as the reference frame, its location in S' is always constant, so $x'_C = x'_D$ (and we will keep calling this x'_C , since conveniently C stands for “car”). Note that by setting the origin at A, the location of the car is *not* $x' = 0$, since the car is not at the location of the eastern person when they clap. Finally, $x_B = x_C = -L$ since the western person is a distance L to the left of the eastern person, and $t_A = t_B = 0$ since their claps are simultaneous in the ground frame and we have set the origin of time at event A. Filling in our table, we now have

Label	Event	Coords in S	Coords in S'
A	eastern person claps	$(0, 0)$	$(0, 0)$
B	western person claps	$(0, -L)$	(t'_B, x'_B)
C	car passes western person	$(t_C, -L)$	$(0, x'_C)$
D	car passes tree	(t_D, x_D)	(t'_B, x'_C)

We have 6 unknowns (t'_B , x'_B , t_C , x'_C , t_D , and x_D) and 6 Lorentz transformation equations relating the coordinates of events B, C, and D, so we have enough information to solve the problem.

Using the time Lorentz transformation from S' to S on event B, we have $t'_B = \gamma(0 - v(-L)/c^2) = \gamma vL/c^2$. Similarly, using the space Lorentz transformation from S to S' on event C, we have $-L = \gamma(x'_C + v(0))$, so $x'_C = -L/\gamma$. These are the only two pieces of information we need to solve for x_D using the space Lorentz transformation from S to S' on event D:

$$x_D = \gamma(x'_C + vt'_B) = \gamma\left(-\frac{L}{\gamma} + \frac{\gamma v^2 L}{c^2}\right) = L\left(-1 + \frac{v^2}{c^2 - v^2}\right) = L\left(\frac{2v^2 - c^2}{c^2 - v^2}\right).$$

When $v = c/\sqrt{2}$, the numerator vanishes, which means that $x_D = 0$, and the tree is at the same location as the eastern person. For $v > c/\sqrt{2}$, $x_D > 0$, which means the tree is to the right (east) of the eastern person. Plugging in the numbers for $v = 4c/5$, we have

$$x_D = L\left(\frac{2(16/25) - 1}{1 - 16/25}\right) = \frac{7}{9}L$$

so the tree is $7L/9$ to the right of the eastern person, or $16L/9$ to the right of the western person.

- (b) (20 points) You should find that your solution to part (a) is only a function of v^2 ; in other words, the tree is at the same position whether or not the car is traveling eastward with velocity v or westward with velocity $-v$. Explain this fact in a few sentences, using a diagram. *Hint: if the direction of the car is reversed, who claps first?*

If the car is traveling westward with velocity $-v$, the western person claps first in S' , as we can see from the fact that $t'_B < 0$ when $v < 0$. This is the relativity of simultaneity at work! The tree is at the same location because the order of events is reversed, so it's like watching a movie backwards: first the car passes the tree when the western person claps, then the car passes the western person when the eastern person claps.

2. **Derivation of velocity addition.** (40 points) Morin exercise 2.17 (p. 106). By solving this problem, you will have completed the derivation of the velocity addition formula you first saw in Lecture 1! (You will see in Lecture 5 how to set up this problem using matrix multiplication. But you don't need that bit of technology to do the homework.)

As computed in lecture, the matrix which takes S to S'' is

$$\Lambda_{v_2}\Lambda_{v_1} = \gamma_1\gamma_2 \begin{pmatrix} 1 + \beta_1\beta_2 & \beta_1 + \beta_2 \\ \beta_1 + \beta_2 & 1 + \beta_1\beta_2 \end{pmatrix}$$

We want to match this to the form $\begin{pmatrix} \tilde{\gamma} & \tilde{\gamma}\tilde{\beta} \\ \tilde{\gamma}\tilde{\beta} & \tilde{\gamma} \end{pmatrix}$, where $\tilde{\gamma}$ is the Lorentz factor associated to the velocity-addition result $\tilde{\beta} = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}$. We know we're on the right track because the diagonal elements should be equal to each other, the off-diagonal elements should be equal to each other, and the ratio of off-diagonal to diagonal elements should be $\tilde{\beta}$, all of which are true.

Now we just need to compute

$$\tilde{\gamma} = \frac{1}{\sqrt{1 - \tilde{\beta}^2}} = \frac{1}{\sqrt{1 - \left(\frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}\right)^2}} = \frac{1 + \beta_1 \beta_2}{\sqrt{(1 - \beta_1^2)(1 - \beta_2^2)}} = \gamma_1 \gamma_2 (1 + \beta_1 \beta_2).$$

This is exactly the diagonal entry of $\Lambda_{v_2} \Lambda_{v_1}$, so we're done!